Incorporating Spatial Correlogram into Bag-of-Features Model for Scene Categorization

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Abstract. This paper presents a novel approach to represent the codebook vocabulary in bag-of-features model for scene categorization. Traditional bag-of-features model describes an image as a histogram of the occurrence rate of codebook vocabulary. In our approach, spatial correlogram between codewords is incorporated to approximate the local geometric information. This works by augmenting the traditional vocabulary histogram with the distance distribution of pairwise interest regions. We also combine this correlogram representation with spatial pyramid matching to describe both local and global geometric correspondences. Experimental results show that correlogram representation can outperform the histogram scheme for bag-of-features model, and the combination with spatial pyramid matching improves effectiveness for categorization.

1 Introduction

In this paper, we consider the representation of visual information, a fundamental problem of automatic scene categorization. A good representation should be discriminative to inter-category variations and robust to intra-category variations. Given a collection of images, a common way is to represent images by color, edge, texture or shape, and then classify them to predefined categories, e.g., street, suburb, bedroom, etc. These representations are also useful in content-based multimedia analysis and retrieval applications [1][2][3].

A popular recognition scheme is to represent and match scenes on the bag-of-features model [4][5]. An image is represented as an orderless collection of local features. SIFT feature [6] is widely used for its good performance in the classification/categorization task. Once local features are obtained, an unsupervised clustering algorithm, typically k-means [7], is applied to quantize the continuous high-dimensional local feature space into a discrete vocabulary of codebook. The bag-of-features model corresponds to a histogram of the occurrence rate of particular vocabulary in the given image.

One drawback of traditional bag-of-features model is the limited descriptive ability of spatial information because orderless feature collection does not contain spatial layout or geometric correspondence of the features. Recent works in bag-of-features model with spatial layout show the use of spatial information

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can improve the classification/categorization quality [8][9][10][11][12][13]. Specifically, a sequence of grids at different resolutions can be constructed for a given image with each grid being represented as a histogram of vocabulary. The histogram with pyramid match kernel [14] is used as features in spatial matching scheme. Approaches based on the spatial pyramid matching achieve significantly improved performance on scene categorization tasks [8]. Another strategy to overcome this drawback is to augment the traditional bag-of-features models with geometric information such as the pairwise relationship between neighboring local features [15][16][17]. However, the computational cost of these methods is high.

We propose a new spatial correlogram approach to represent geometric information and improve bag-of-features model for visual categorization. Informally, the spatial correlogram of two codewords v_i and v_j in vocabulary indicates the distance distribution of the positions of interest regions, which are encoded to v_i and v_j, respectively. The highlights of this representation are: (i) it contains the local geometric correspondence; (ii) with similar storage and computational cost, the categorization accuracy of correlogram representation outperforms the histogram in bag-of-feature model.

The remainder of this paper is organized as follows. Section 2 reviews the related literature on bag-of-features model. Section 3 presents our proposed spatial correlogram approach for scene categorization. Section 4 discusses the experimental setup. In Sect. 5 we present our experimental results. And finally, we conclude our work in Sect. 6.

2 Related Work

Bag-of-features model has been studied for video retrieval [4] and image categorization [5]. Local features of an image are extracted and represented as a codeword according to vocabulary of codebook. The histogram of vocabulary is generated to describe the image. There are many attempts to incorporate the spatial layout or geometric correspondence with the basic bag-of-features models [13][8]. Specifically, Fei-Fei and Perona [13] used a bayesian hierarchical model to learn natural scene categories. The algorithm [13] was modified based on Latent Dirichlet Allocation model(LDA) [18] and proposed to represent codebook distribution of each category. Another improvement on bag-of-features model focuses on the spatial layout information. Lazebnik et al. [8] introduced spatial pyramid matching for scene categorization based on global geometric correspondence. The image is partitioned into subregions and a histogram of local features is calculated inside each subregion. The experiments showed excellent performance on three diverse datasets.

Besides the global geometric correspondence, a more intuitive and efficient approach is to add pairwise relations of neighboring local features [4][16]. Sivic and Zisserman [4] considered spatial consistency, which is measured by requiring that neighboring matches in the query region lie in a surrounding area in the retrieved frame. In [16], “doublets”, a second vocabulary built on spatial co-occurrences

of word pairs, is included to encode spatially local co-occurring regions for image segmentation. However, these methods lead to large computational costs. The focus of this paper is to propose a new representation which contains sufficient spatial co-occurrence information of codebook vocabulary, and computes with efficiency.

3 Spatial Correlogram Approach

Histogram is a simple and powerful representation as a method for image description in computer vision. In bag-of-features model, an image is firstly represented as an orderless collection of local features. Given a set of training images, an unsupervised clustering algorithm, typically k-means, is applied to generate the codebook vocabulary $V = \{v_1, v_2, \cdots, v_n\}$, where n is the size of codebook. A codebook histogram is obtained for each image with each bin representing a codeword v_i in V:

$$Hist(v_i) = \sum_{k=1}^{N} \begin{cases} 1, \text{ if } v_i = \arg\min_{v \in V}(Dist(v, r_k)) \\ 0, \text{ otherwise} \end{cases} \tag{1}$$

where r_k denotes the kth interest region, N is the total number of interest regions in the image, and $Dist(v, r_k)$ is the distance between vocabulary v and interest region r_k in local feature space. The histogram approximates to the global distribution of vocabulary frequency.

One drawback of histogram representation is its limited descriptive ability of spatial information. In image indexing/retrieval area, Huang et al. [19] proposed the color correlogram feature to distil the spatial correlation of colors. This feature achieves better performance than the traditional color histogram scheme. Our strategy is inspired based on the color correlogram approach to improve the bag-of-features model. We differ from their representation [19] by incorporating "neighboring" codewords in vocabulary to reduce the dimension of representing features.

In a given image $\mathcal{I}$, suppose $p_k = (x_k, y_k)$ is the position of interset region r_k, and the corresponding codeword of r_k in the codebook is $v(r_k)$. A table indexed by codebook vocabulary and position distance in image is used to describe the distance distribution between interest region r_k and its neighbors:

$$T_k(v_j, d) = \frac{|\{r_i \in \mathcal{I} | v(r_i) = v_j, \|p_k - p_i\| = d\}|}{|\{r_i \in \mathcal{I} | v(r_i) = v_j\}|} \tag{2}$$

where $r_i \in \mathcal{I}$ means interest region r_i in image $\mathcal{I}$, and $\|p_k - p_j\|$ indicates the position distance between interest region r_k and r_j. For convenience, L_∞ norm is used to measure this distance, i.e., $\|p_k - p_j\| = \max\{|x_k - x_j|, |y_k - y_j|\}$. To describe the general distribution from codeword v_i to v_j over the whole image, we use the following equation:

$$T(v_i, v_j, d) = \sum_{v(r_k) = v_i} T_k(v_j, d) \tag{3}$$

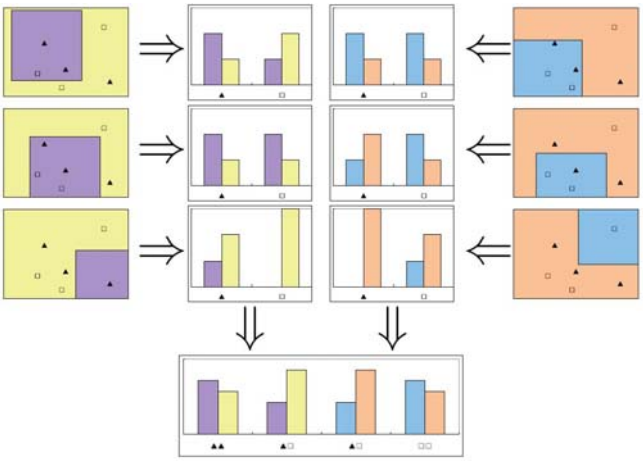


Fig. 1. Toy example of Corr scheme in an image with codebook vocabulary $V = \{\blacktriangle, \square\}$ and step $s = 2$. At the top, the Corr table indexed by vocabulary and step is calculated for each interest region. Next, accumulate the table whose interest regions map the same codeword and generate the representation.

Comparing with histogram representation(1), $T(v_i, v_j, d)$ includes information about distance distribution of the position of codewords in codebook vocabulary. However, the storage and computational requirement is larger than histogram. Assume the image size is $w \times h$, the potential maximum position distance is $D = \max(w, h)$, and the vocabulary size is n. $O(n^2 D)$ space is required while $O(n)$ is required for histogram representation.

Two phases are considered in order to reduce the storage and computational cost. Instead of using position distance directly to describe the pairwise relations, we divide the range from 0 to D into s "steps"; each step is represented by a distance interval $\mathbf{I} = [a, b)$. For all examples in this paper, we segment the range $[0, D)$ as $\mathbf{I}_1 = [0, L)$, $\mathbf{I}_2 = [L, 2L)$, $\ldots$, $\mathbf{I}_s = [(s-1)L, D)$ where $L = \frac{\min(w,h)}{s}$. Equation (3) can be rewritten as:

$$Corr(v_i, v_j, \mathbf{I}) = \sum_{d \in \mathbf{I}} T(v_i, v_j, d) = \sum_{d \in \mathbf{I}} \sum_{v(r_k) = v_i} T_k(v_j, d) \tag{4}$$

In order to distinguish this correlogram scheme and its variants in the following part, we call Equ. 4 as "Corr" scheme, while correlogram approach represent both Corr and its variants. The storage space of $Corr(v_i, v_j, \mathbf{I})$ is reduced to $O(n^2 s)$. Moreover, Equ. 4 describes the spatial correlogram of a flexible range and avoids the overfitting problem. An illustration of Corr scheme is shown in Fig. 1.

Furthermore, we exploit the discriminative ability of single pairwise spatial correlogram. A simple and intuitive way is to capture spatial correlation between

identical codeword only. This self-correlogram approach(Self-Corr) is defined according to the following formula:

$$Self_Corr(v_i, \mathbf{I}) = Corr(v_i, v_i, \mathbf{I}) = \sum_{d \in \mathbf{I}} \sum_{v(r_k)=v_i} T_k(v_i, d) \tag{5}$$

The corresponding space of Self-Corr is $O(ns)$, depending on the number of steps s. In fact, Self-Corr is degenerated to the histogram representation when $s = 1$.

Besides the Self-Corr approach, we augment the correlation by examining the vocabulary and its neighborhood in the original local feature space; i.e., for each vocabulary v_i, a subset of neighboring codewords V_i in codebook is also found using k-nearest neighbor(kNN) algorithm, and only the codeword $v_j \in V_i \cap \{v_i\}$ is used to compute the spatial correlogram $Corr(v_i, v_j, \mathbf{I})$ in Equ. 4 which is called as kNN-Corr. In our experiments, we choose k=1 or 3. This strategy is a trade-off between Corr approach(4) and Self-Corr approach(5): the storage and computational cost is lower than Corr approach if the value k is not large, and it is able to represent more spatial information of distance distribution than the Self-Corr approach. Comparison of these approaches will be discussed in Sect. 5.

4 Experimental Setup

We evaluate the correlogram approach on a scene dataset from Lazebnik et al. [8]. There are 15 natural scene categories and 200 to 400 images per category. For each category, 100 images are randomly selected for training and the rest for testing. We repeat this random operation 10 times and generate 10 pairs of the training/testing subset. Sample images of the dataset are shown in Fig. 2.

The implementation of codebook follows the general scheme for scene categorization based on the bag-of-features model. For each image in the dataset, a set of local features are firstly extracted. We follow the image feature setting from [8][10]. Dense sampling over image grids is then applied because of the advantage for learning scene categories reported in [13]. In our experiment, SIFT descriptor [6] is used to describe the interest regions for its good performance for object and scene recognition [21][22]. To generate the codebook vocabulary, k-means clustering is selected since it is simple and widely used in previous works [4][5][8][13][22]. The performance difference between k-means and other clustering algorithm are not the focus of this paper. The vocabulary sizes we use are {16, 25, 50, 100, 200, 400, 800, 1600}. We define the representations with less than or equal to 100 codewords as "weak feature" and otherwise as "strong feature" for the different discriminative power.

To get the categorization accuracy, support vector machine(SVM) with histogram intersection kernel is used in our experiment. Specifically, libSVM [23] is used to train using the one-versus-one rule for multi-class classification. The categorization rate in the following section is the average of categorization accuracy over the ten random selected testing subsets.

Fig. 2. Sample images of the 15 natural scene categories. Eight of these (a-h) were originally collected by Oliva and Torralba [20], five (i-m) by Fei-Fei and Perona [13], and two (n-o) by Lazebnik et al. [8].

5 Experiments and Results

In this section, we compare correlogram approach with the traditional histogram representation and show the results based on the 15 natural scene categories.

5.1 Experiment 1: Comparison of Correlogram and Histogram

We start the experiment with the comparison of correlogram and histogram representation. As introduced in Sect. 3, we employ five kinds of representation strategies for weak features: histogram(1), Self-Corr(5), 1NN-Corr, 3NN-Corr, and Corr(4). For the four kinds of correlogram representations, we set the position distance separated into 2 steps, i.e., two distance intervals for each image. The storage and computational cost increase from histogram to Corr scheme. The experimental results are shown in Fig. 3.

For weak feature, we can see that Corr scheme performs the best among all the representations. From the baseline histogram scheme, the average categorization accuracy improves at the rate of 5.5%-7.4% to Self-Corr approach, and 9.8%-15.3% to Corr approach scheme. Specifically, the Corr approach with 100 vocabulary achieves the accuracy 71.42%, which is comparable to the histogram scheme with 1600 vocabulary. However, the storage and computational cost of the Corr representation are extremely expensive.

For strong features, we only investigate the Self-Corr, 1NN-Corr, and 3NN-Corr schemes for efficiency. Though the difference between results of histogram and Self-Corr scheme decreases while the codebook size increase, the Self-Corr

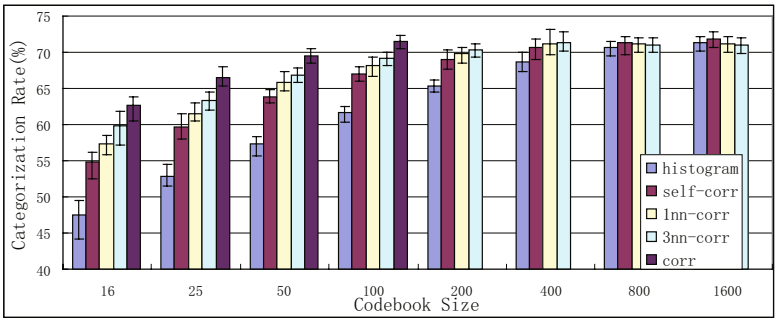


Fig. 3. Evaluation of the bag-of-features model with histogram and correlogram representation strategies for both weak features and strong features. The codebook size $|V| \in \{16, 25, 50, 100, 200, 400, 800, 1600\}$.

Table 1. Histogram and Self-Corr scheme with same representation dimension

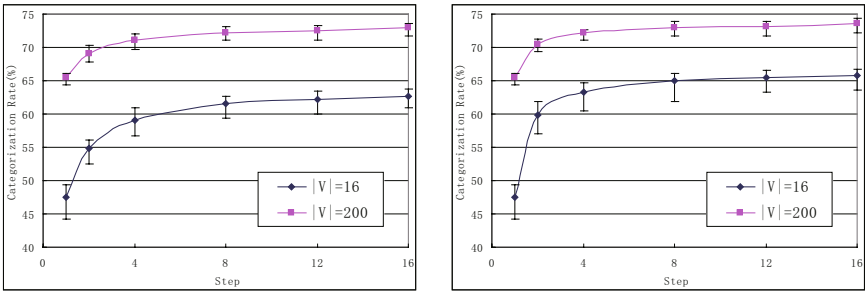
Repr. dim.	50	100	200	400	800	1600
Histogram	57.4	61.6	65.4	68.7	70.6	71.3
Self-Corr	59.7	63.9	67.1	69.0	70.6	71.4

can still outperform the histogram method with the same codebook size. For 1NN-Corr and 3NN-Corr schemes, the categorization accuracy performs better than that of Self-Corr before the codebook size increases to 800; when the codebook size is larger than 800, the performance is not as good as Self-Corr.

It is also interesting to compare performances of different representations with same storage space. Using a codebook with vocabulary size n, histogram and Self-Corr need $O(n)$ and $O(2n)$ space in this experiment, respectively. We compare histogram and Self-Corr scheme with the same representation dimension, e.g., histogram with $n = 50$ versus Self-Corr with $n = 25$, histogram with $n = 100$ versus Self-Corr with $n = 50$, etc. Generally, codebook with larger size has higher classification accuracy. However, Table 1 lists the comparison results: with same representation dimension, Self-Corr representations perform as well as or better than histogram; that means, incorporating spatial correlogram information to bag-of-features model is comparable with the histogram scheme even though the vocabulary size for correlogram method is half of which for histogram.

5.2 Experiment 2: Step in Correlogram Representation

In this subsection we focus on the impact of different step values, which is an important factor in our correlogram representation. To explore the role of step value in correlogram representation, we experiment different step $s \in \{1, 2, 4, 8, 12, 16\}$ with Self-Corr and kNN-Corr ($k = 3$) schemes based on weak feature ($|V| = 16$) and strong feature($|V| = 200$). Note that step $s = 1$ indicates the baseline histogram representation. The results from Fig. 4 show that increasing the step



(a) Self-Corr (b) 3NN-Corr

Fig. 4. Correlogram schemes with different steps

Table 2. Categorization results of the bag-of-features model and the spatial pyramid with histogram and correlogram representations. For completeness, both single-level(the "single" columns) and multi-level(the "pyramid" columns) pyramids are listed in the table.

	Weak feature($\|V\| = 16$)				Strong feature($\|V\| = 200$)			
Repr.	Histogram		Correlogram		Histogram		Correlogram	
Level	Single	Pyramid	Single	Pyramid	Single	Pyramid	Single	Pyramid
0	47.4	-	54.8	-	65.4	-	69.0	-
1	56.7	58.5	61.8	66.0	71.5	71.9	71.7	73.3
2	60.2	**62.5**	62.6	**67.4**	72.5	**72.8**	71.3	**74.0**

clearly improves over the correlogram representation for both weak features and strong features. Specifically, the categorization accuracy increases fast when step value grows from 2 to 8. Note that increasing step value from $s = 8$ to $s = 16$ results in a small performance increase for all the curves in Fig. 4, probably due to the limitation when the distance intervals are too small and unable to capture the general distance distribution of neighborhood.

5.3 Experiment 3: Incorporating with Spatial Pyramid

In particular, we consider incorporating correlogram approach with the spatial pyramid matching scheme [8]. Images are partitioned into subregions. The histogram and Self-Corr(with step $s = 2$) of codebook are found inside each subregion. The maximum level of our experiment is 2, i.e., images on 1×1, 2×2 and 4×4 grids. From the results shown in Table 2, it is observed that the multi-level spatial pyramid matching performs better than the single-level bag-of-features model for both correlogram and histogram representation.

For weak feature, the histogram representation improves from single-level to multi-level pyramid at the rate of 1.8% for level 1 and 2.3% for level 2, while in Self-Corr representation the improvement rates are 4.2% and 4.8%, respectively,

which is better than histogram representation. The result shows the effectiveness of combining Self-Corr and spatial pyramid matching for weak feature.

For strong feature, we find that the method fails to improve single-level categorization rate when the level grows from 1 to 2 for strong feature's Self-Corr scheme, probably because images are over-subdivided under 4×4 grids for correlogram. However, the multi-level accuracy still increases from 73.3% to 74.0%. And the same multi-level pattern as for weak feature is observed: correlogram performs better than histogram at level 1 and 2.

Another observation is that using the spatial pyramid, Self-Corr at level 1 achieves better performance than histogram at level 2, for weak and strong features. We can see both of spatial pyramid matching and spatial correlogram approximate the geometric correspondence of image. Spatial pyramid matching works by computing at global or subregion scale to represent the spatial layout, and our spatial correlogram represents with pairwise relations between codewords, or alternately the neighboring local features in the original local feature space. Framework combined with these two methods can include global and local geometric correspondence and is able to show powerful descriptive ability of spatial information.

6 Conclusion

In this paper, we propose the spatial correlogram representation to incorporate geometric information and improve the bag-of-features model for scene categorization. The histogram representation for bag-of-features model is replaced by the pairwise relationship of codebook vocabulary. This relation is explicitly extracted from the distance distribution of pairwise interest regions. Two kinds of variants are considered in order to reduce the storage and computational cost: the self-correlogram scheme captures spatial correlation between identical codeword only, and the kNN-correlogram scheme exploits the discriminative ability of the correlation between codeword and its neighborhood in the original local feature space. Experiments on 15 natural scene categories show that the correlogram representation can outperform the traditional histogram scheme with the same codebook size or the same representation dimension in the bag-of-features model. In addition, we examine the combining of correlogram representation and spatial pyramid matching. These methods have different but complementary abilities to express spatial information of an image. The experiments show that the combination achieves powerful descriptive ability of spatial information for scene categorization.

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